Lessons

Word Problems with Measurements

Distance, **rate**, and time are all related. If you travel faster, you can cover more distance in less time. You can travel 200 miles in a lot less time if you drive at 50 miles per hour than if you drive at 20 miles per hour.

The **distance formula** shows the relationship between distance, rate, and time: **distance (d) = rate (r) × time (t),** or $d = rt$

If you know any two of the values in the distance formula, you can find the third by using one of the formulas shown in the following examples.

Example 1

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Rudy drove an average of 50 miles per hour for $3\frac{1}{2}$ hours. How far did he drive?

Step 1

Write the distance formula to find the distance.

$d = rt$

Step 2

Substitute the known values for rate and time.

$d = 50 \times 3\frac{1}{2}$

Step 3

Multiply to find the distance.

$\frac{\cancel{50}^{25}}{1} \times \frac{7}{\cancel{2}_{1}} = 175$ miles

Rudy drove **175 miles** in $3\frac{1}{2}$ hours at 50 miles per hour.

Example 2

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Theresa made the trip to her mother's house in $2\frac{1}{2}$ hours. If the distance she drove is 100 miles, what was her average speed (rate)?

Step 1

Rewrite the distance formula to find the rate.

$\frac{d}{t} = r$

Step 2

Substitute the known values for distance and time.

$r = 100 \div 2\frac{1}{2}$

Step 3

Divide to find the rate.

$\frac{\cancel{100}^{20}}{1} \times \frac{2}{\cancel{5}_{1}} = 40$ miles per hour

Theresa drove at an average speed of **40 miles per hour.**

Example 3

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How long will it take Henri to travel 120 miles if he drives at an average speed of 40 miles per hour?

Step 1

Rewrite the distance formula to find the time.

$\frac{d}{r} = t$

Step 2

Substitute the known values for distance and rate.

$t = 120 \div 40$

Step 3

Divide to find the time.

$40\overline{)120} = 3$ hours

It will take Henri **3 hours** to travel 120 miles at 40 miles per hour.

Skill Tip

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When using a formula, it is often helpful to write the formula out as the first step in your work. Then, identify all of the known quantities given to you in the problem and substitute them into the formula.

You may need to rewrite the formula in a different way to solve the problem. Notice that in Example 2, the formula $d = rt$ is written as $\frac{d}{t} = r$.

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Lesson 1: Customary Units of Measurement

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